# Smart Lamppost — Simultaneous Sensor Conflict Resolution Logic

## SECTION A — DETECTION PRIORITY HIERARCHY (highest wins in conflict)

| Priority Rank | Scenario / Conflict | Winning Behavior | PWM Output | Decay Timer | Energy Mode | Sensor(s) Governing Decision | Rationale |
|---|---|---|---|---|---|---|---|
| **1 — HIGHEST** | Vehicle + Pedestrian detected simultaneously in same zone | Vehicle rules: full 100% brightness | 255 (100%) | 3–5 s post-vehicle departure | Peak Mode | Microwave Radar + Edge Camera (vehicle bounding box dominates) | Vehicle at speed poses the greatest safety risk. Pedestrian is also safer under full illumination. No conflict — brightest mode serves both. |
| **2** | Pedestrian detected; ambient ALS reads sudden brightness drop (storm/cloud) | Pedestrian floor raised to 60% (override +10%) | ~153 (60%) | Continuous while both conditions hold | Elevated Safety Mode | PIR + ALS (BH1750 lux threshold < 20 lux triggers uplift) | Low visibility atmospheric event elevates pedestrian safety floor above standard 40–50% baseline. |
| **3** | RTC midnight curfew active + Vehicle approaches | Vehicle capped at 50% brightness (curfew override) | ~128 (50%) | 3–5 s post-departure, decays to 10% | Curfew Mode | DS3231 RTC gate overrides camera/radar PWM ceiling | Sky glow mitigation at night overrides full-beam. 50% still provides adequate road visibility per IES RP-8 guidance. |
| **4** | RTC midnight curfew active + Pedestrian detected | Pedestrian at 35% (curfew-adjusted floor) | ~89 (35%) | Continuous while pedestrian in zone | Curfew-Pedestrian Mode | DS3231 RTC + PIR + Edge Camera | Slightly lower floor than daytime standard; preserves pedestrian safety while respecting noise/skyglow ordinances. |
| **5** | False trigger (swaying tree / animal) + No human/vehicle class confirmed | No change — hold 10% eco-baseline | ~26 (10%) | N/A | Eco Mode | TinyML edge classifier rejects non-human/vehicle spectral signatures | Prevents ecosystem disruption and unnecessary actuations. AI confidence threshold must exceed 85% for any PWM change. |
| **6 — LOWEST** | All sensors idle, no detection, daytime ALS confirms high ambient lux | Full off (or 0%) — daytime suppression | 0 | N/A | Daytime Sleep | ALS (lux > 500) + RTC (within daylight window) | Dual-confirmation prevents false daylight suppression from e.g. a single car headlight sweep. |

## SECTION B — EDGE-CASE DECISION TABLE

| Edge Case | Trigger Condition | Sensor State | System Response | PWM | Notes / Risk Flag |
|---|---|---|---|---|---|
| Sensor dropout mid-sequence | Edge camera FPS drops to 0 mid-vehicle pass (CPU overload) | Radar still active; camera offline | Radar alone holds 100% until dropout clears; camera resumes classification | 255 | ⚠ Radar-only mode has no class confirmation — risk of false hold. Watchdog timer should force decay after 10 s without camera ping. |
| Contradictory classification | Radar detects high-velocity object; camera classifies 'pedestrian' | Fusion mismatch | Default to higher-safety ruling: treat as vehicle (100%) until next camera frame resolves class | 255 | Occurs if cyclist in blind spot. Acceptable failure mode — over-illumination is safer than under. |
| PIR saturated (multiple pedestrians) | Group of >5 pedestrians causes continuous PIR signal | PIR = high, camera = multiple 'person' bounding boxes | Hold 50% brightness; no escalation beyond pedestrian ceiling | ~128 | Group scenario doesn't require higher illumination; 50% is sufficient per safety baselines. |
| ALS sensor failure (stuck value) | ALS returns constant 0 lux despite daytime | ALS = 0, RTC = daytime window | RTC takes precedence; daytime suppression still applied. ALS fault logged. | 0 (daytime) | ⚠ Flag for maintenance. Dual-sensor redundancy prevents runaway full-night illumination. |
| Power brownout recovery | MCU reboots mid-session | All sensors reinitialising | Default to 30% safe-startup brightness for 5 s while sensors calibrate | ~77 (30%) | Prevents both total darkness on reboot and full-beam flash on power recovery. |